

Perquinolone Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-334/A

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. No changes to the approved analytical procedures are proposed in this submission.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. No changes to the approved analytical procedures are proposed in this submission.

Section 3.2.P.4 — Excipients

For the commercial formulation F-334/A, Avicel PH-102 (Microcrystalline cellulose, filler-binder) is used.

Selection Justification

The basis for selection of Avicel PH-102 is consistent dissolution performance across registration batches, demonstrated across three pilot-scale batches.

The excipient control strategy was developed in accordance with ICH Q8(R2), ICH Q3D and Ph. Eur. general monograph 2034.

Current Commercial Specification

The current approved commercial specification (SPEC-EX-3320 Rev 05) lists Microcrystalline cellulose as Avicel PH-102.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

The control strategy links critical quality attributes to critical process parameters identified during development. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Deviations observed during the campaign were investigated and closed with no impact to product quality. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions.

The control strategy links critical quality attributes to critical process parameters identified during development. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. No changes to the approved analytical procedures are proposed in this submission. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. No changes to the approved analytical procedures are proposed in this submission. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.